Module 1: Getting Started

Course Overview

- Every Tuesday : 3 hours lab (starting from Jan 13)
- Every Monday: 1 hour lecture (starting from Jan 12)
- Email: Mahshid.Arastonejad@humber.ca
- Respond within 4-6 business hours

Course Overview

Assessment	Weight
In-Class Participation Group	10%
Discussion Assignments Lab	15%
Assignments	50%
Group Report – Clinical Case Study	25%
Total	**%**

Class Participation

- Weekly class poll 1% per
- class * (10 out of 12 classes)
- Week 12 is mandatory attendance

Course Overview

Assessment	Weight
In-Class Participation Group	10%
Discussion Assignments Lab	15%
Assignments	50%
Group Report – Clinical Case Study	25%
Total	**%**

Group Discussion Assignments

- Complete as a group after class
- 3 assignments x 5% Use the
- content to build knowledge towards completing the final group report

Course Overview

Assessment	Weight
In-Class Participation Group	10%
Discussion Assignments Lab	15%
Assignments	50%
Group Report – Clinical Case Study	25%
Total	**%**

Lab Assignments

- 5 lab assignments x 10% Each
- assignment is split into two parts (Part A and B) and due on the Friday 11:59 PM after Part B
- For each assignment, total mark of the assignment = total of Part A + total of Part B.

Course Overview

Assessment	Weight
In-Class Participation Group	10%
Discussion Assignments Lab	15%
Assignments	50%
Group Report – Clinical Case Study	25%
Total	**%**

Group Report - Clinical Case Study

- Run a RNASeq pipeline as a group and write a case study report.
-

Missed and Late Evaluation Policy

- All students requesting an extension must follow the missed and late evaluation policy.
- **(Mandatory)** Complete this form for each missed assignment at least 24h before the assignment deadline.
- Evaluation of missed and late assignments is subjected to my approval.

Class Policy on AI Usage

- Humber College's AI Policy (link) Contends that un-cited and/or other
- unauthorized use of AI in assessments and assignments constitutes academic misconduct as defined in Humber’s Academic Regulations.
- **Do not provide ChatGPT (or any other AI tool) human information that is not open access.**
- **Cite which tool you used and its contribution to your assignment.**
- **You are responsible for the accuracy of your answers.**

Class Policy on AI Usage

Lab Assignments

Code Example:

```
def foo():
    # some code here
    return answer

# Source: My code was generated by claude
```

Short Answer Example:

```
This figure shows that Gene A in fruit fly is upregulated in the treatment group.
Source: I made ChatGPT read the figure.
```

Class Policy on AI Usage

Group Assignments & Group Report

- Communicate with your group members how you used AI tools in your contributions
- Cite this information at the end of your group submission

```
AI Sources: - Member A used ChatGPT to proofread the
Introduction section
```

Learning Strategies

- Coding is a technical skill that requires continuous learning. Software manuals and
- documentations are your friends. Use your creativity to find answers ethically.
- Most knowledge is publicly available.

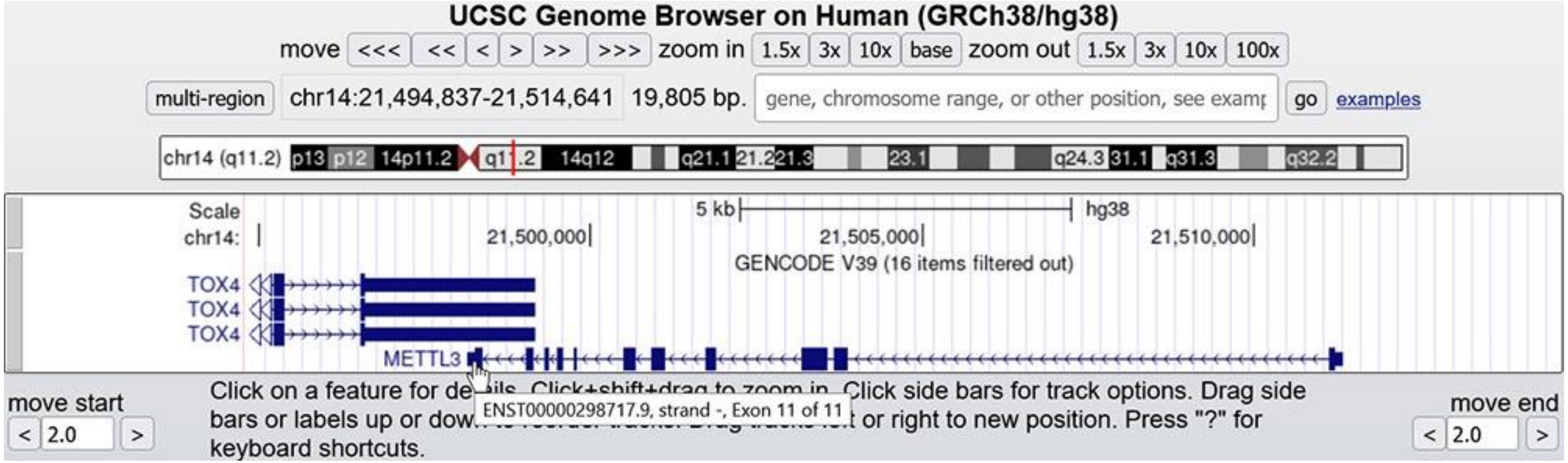
Applications of genomics in clinical medicine [1]

- ***Clinical diagnosis***: Used as diagnostics criteria for medical conditions. ***Disease gene***
- ***identification***: Identify the role of specific genes in heritable diseases. ***Cancer***
- ***genomics***: Used to understand how genomic variants in somatic cells are involved in the initiation and progression of cancer.
- ***Disease treatment***: Inform targeted gene therapies to treat patient monogenic disorders in personalized medicine.
- ***Prenatal diagnosis***: Risk assessment for genetic disorders in pregnancies.

Critical Path

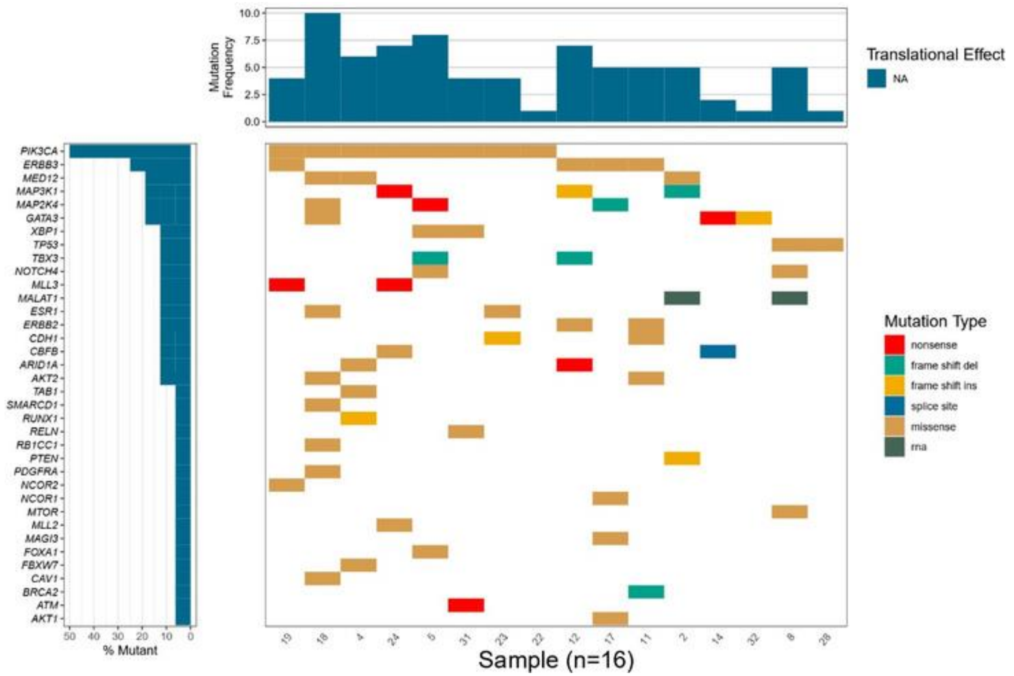
- The lecture materials provide information to support the lab components:
  - Understanding the genetic basis of heritable diseases using public resources
  - RNASeq using Nextflow
  - Gene expression profiling using Nextflow
  - Somatic variant analysis using Nextflow
  - Linkage analysis and clinical applications of GWAS
- Find more details in the Critical Path pdf

Public Databases



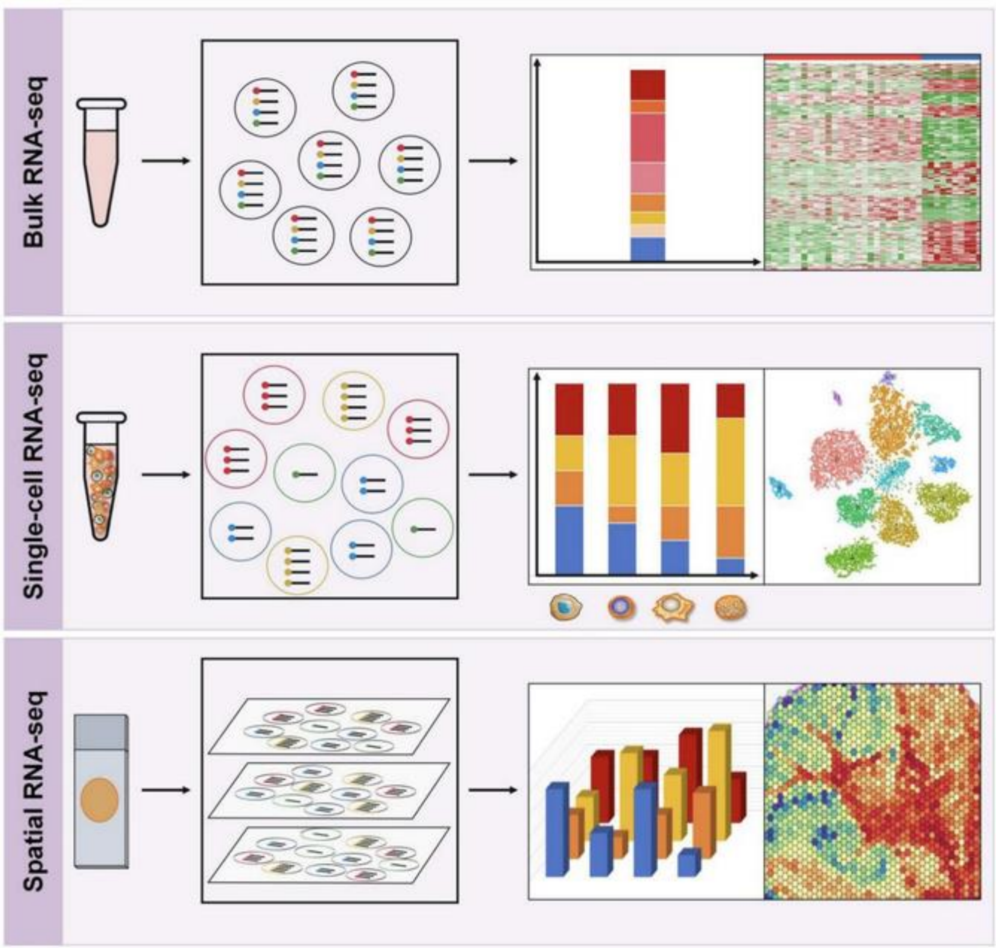
- Public databases that provide biological and medical information to prepare a clinical report.

GenVisR: “Genomic Visualizations in R”



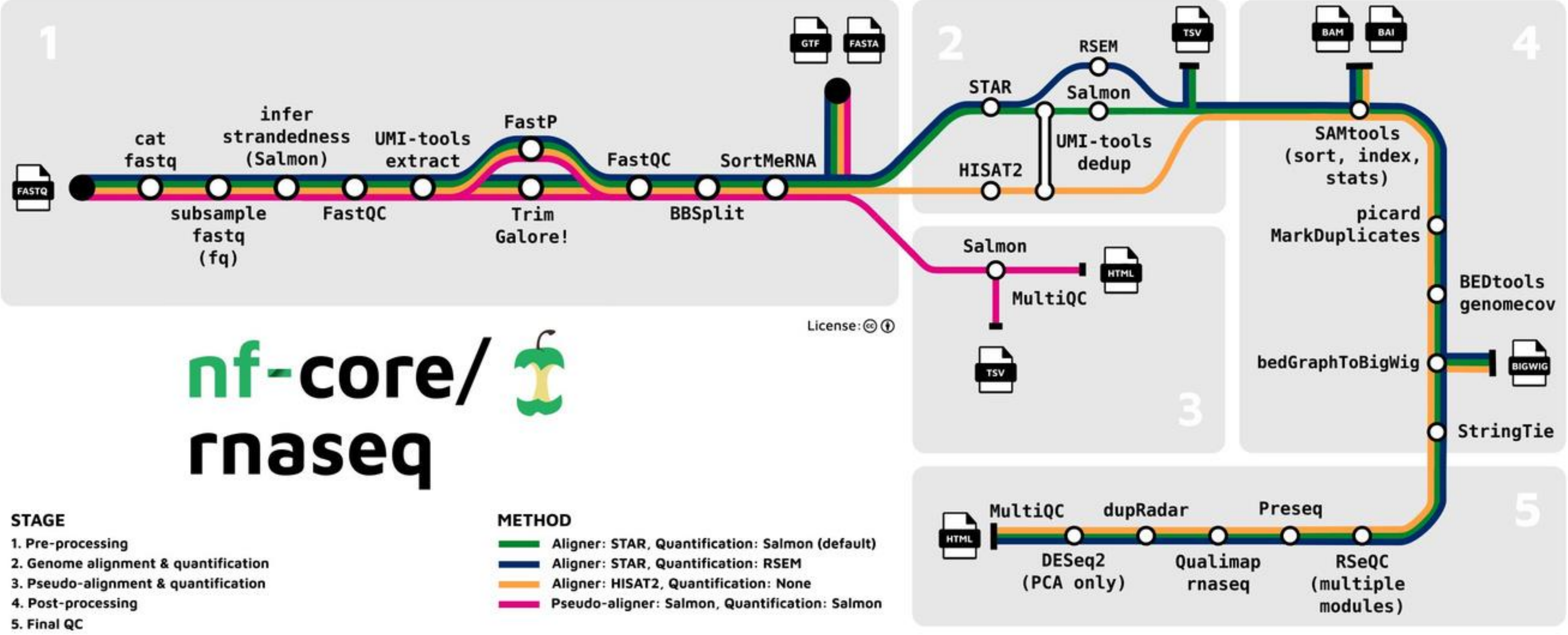
- An R package that creates highly customizable publication-quality graphics supporting multiple species and focused primarily on a cohort level (i.e., multiple samples/patients).

RNA Sequencing 2

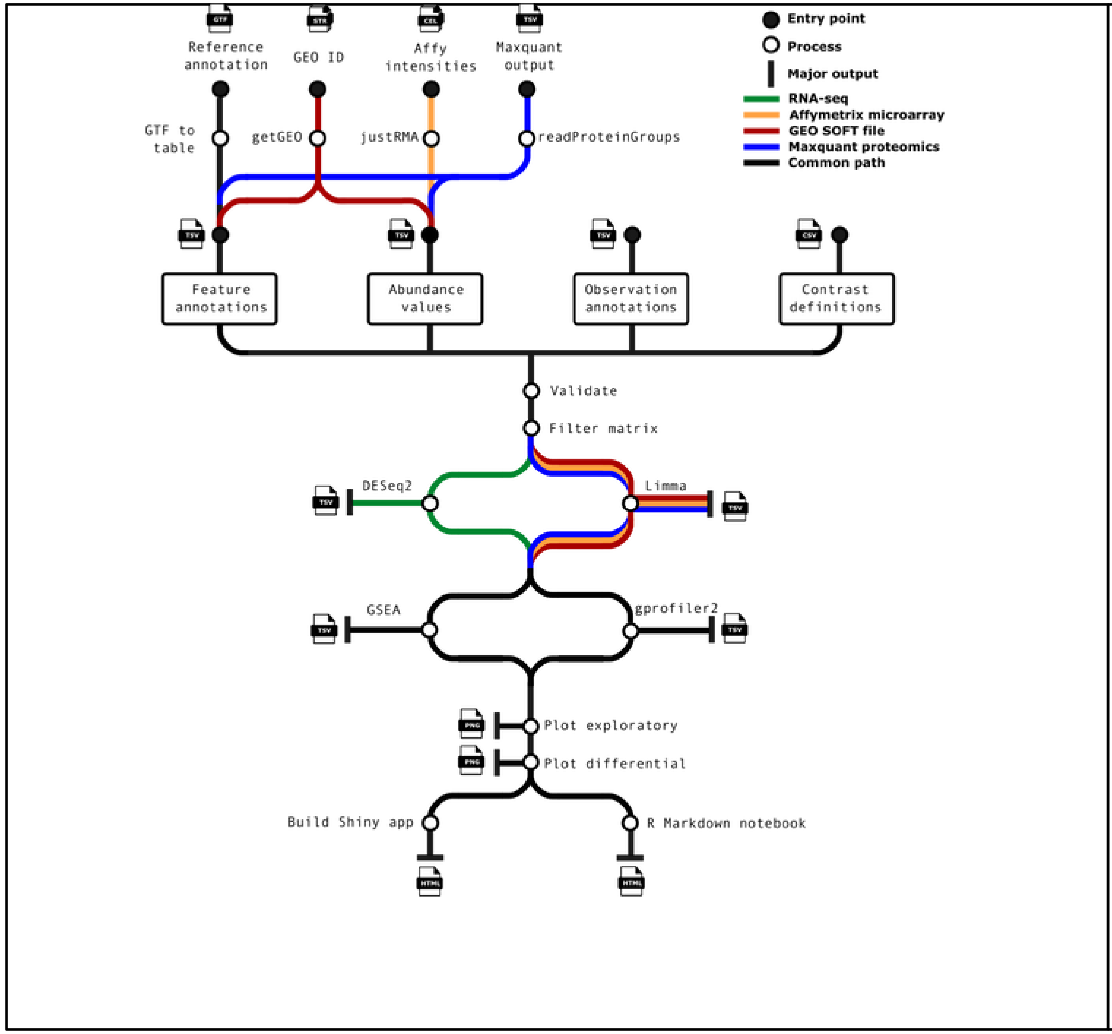


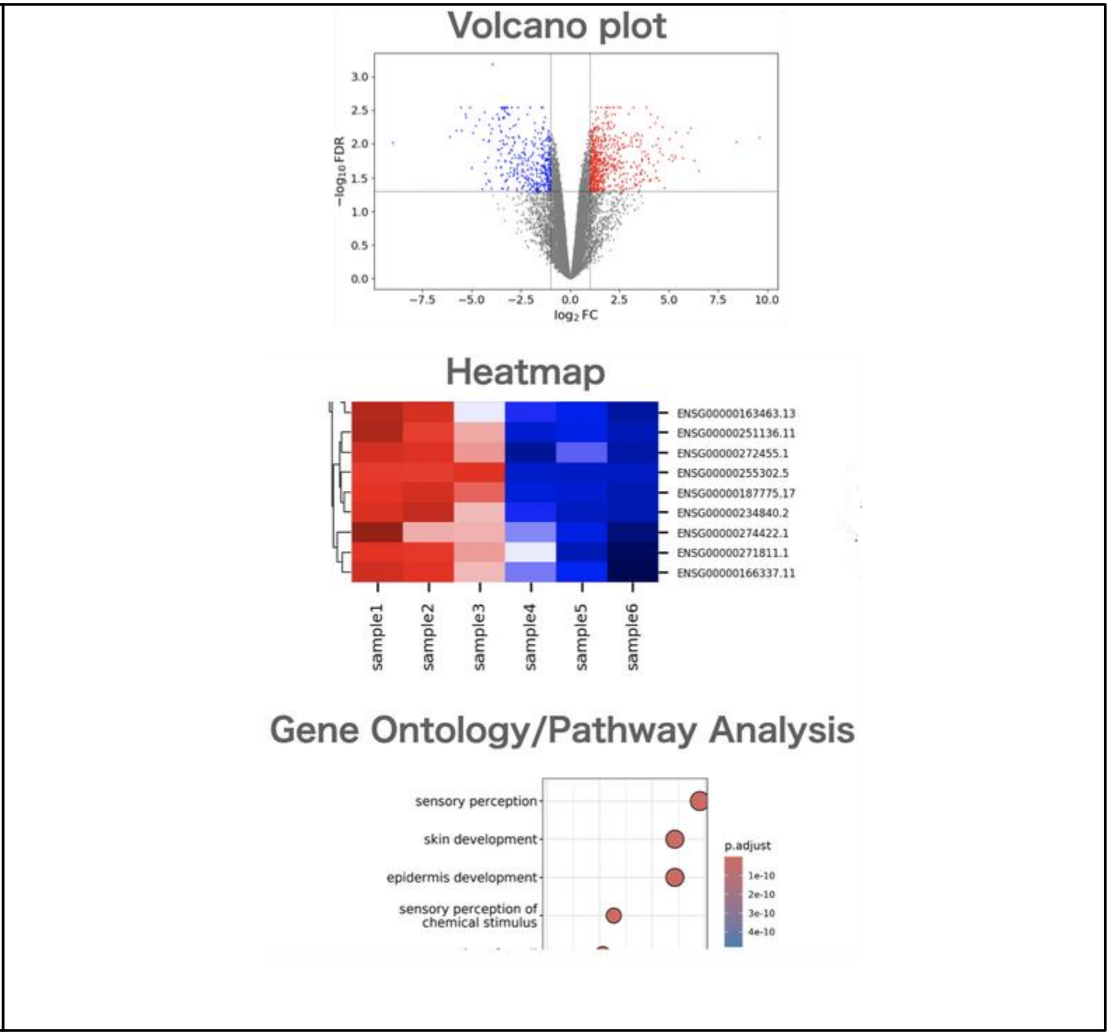
- **Bulk RNA-Seq** provides an average measure of gene expression across the entire population of cells.
- **scRNA-Seq** analyzes gene expression at the single-cell level, which helps to study cellular diversity and identify unique cell types
- **Spatial RNA-Seq** profiles gene expression with spatial resolution in a 3D context within tissue samples.

Bulk RNASeq using Nextflow 3

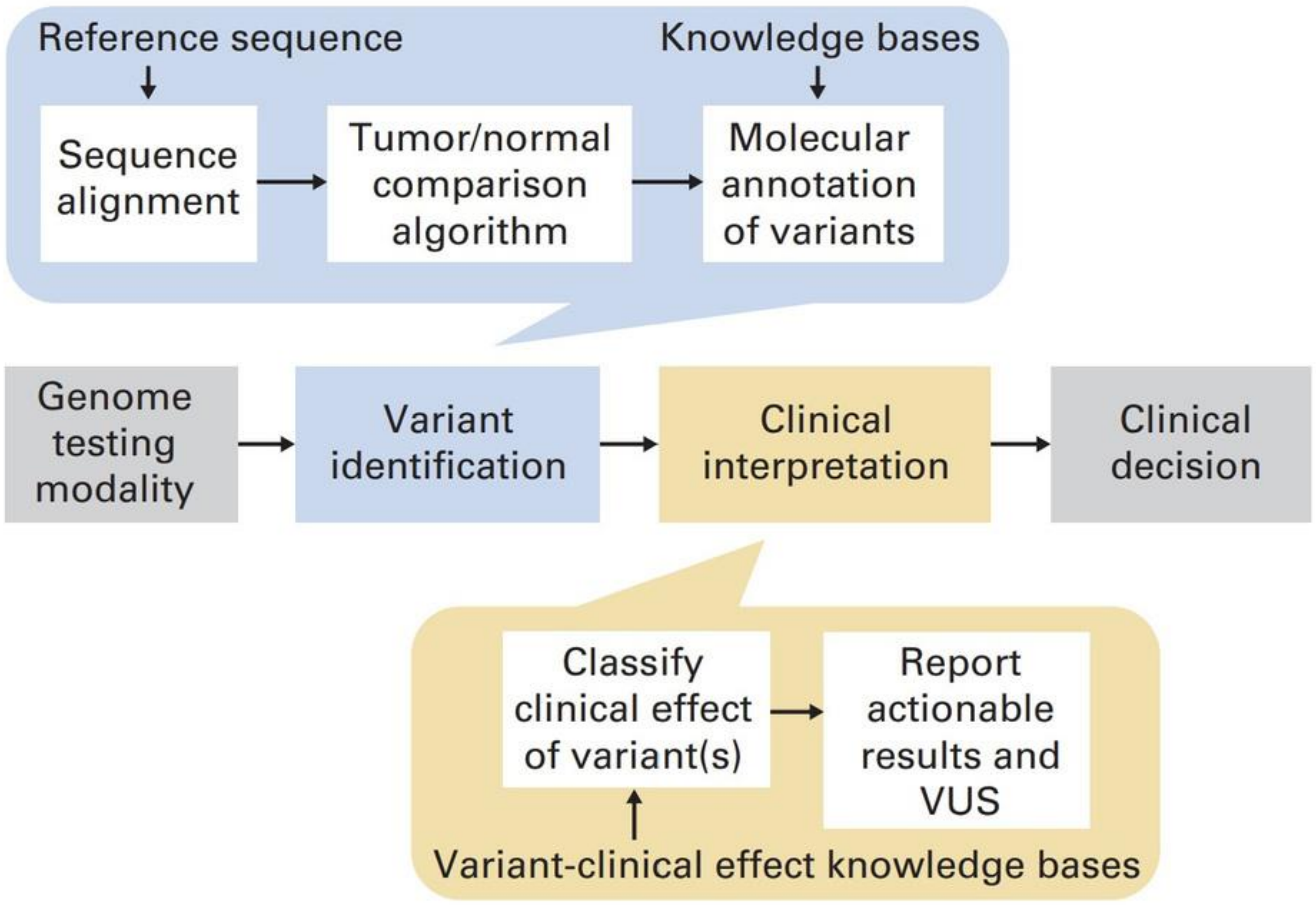


Differential Expression Analysis using Nextflow 4

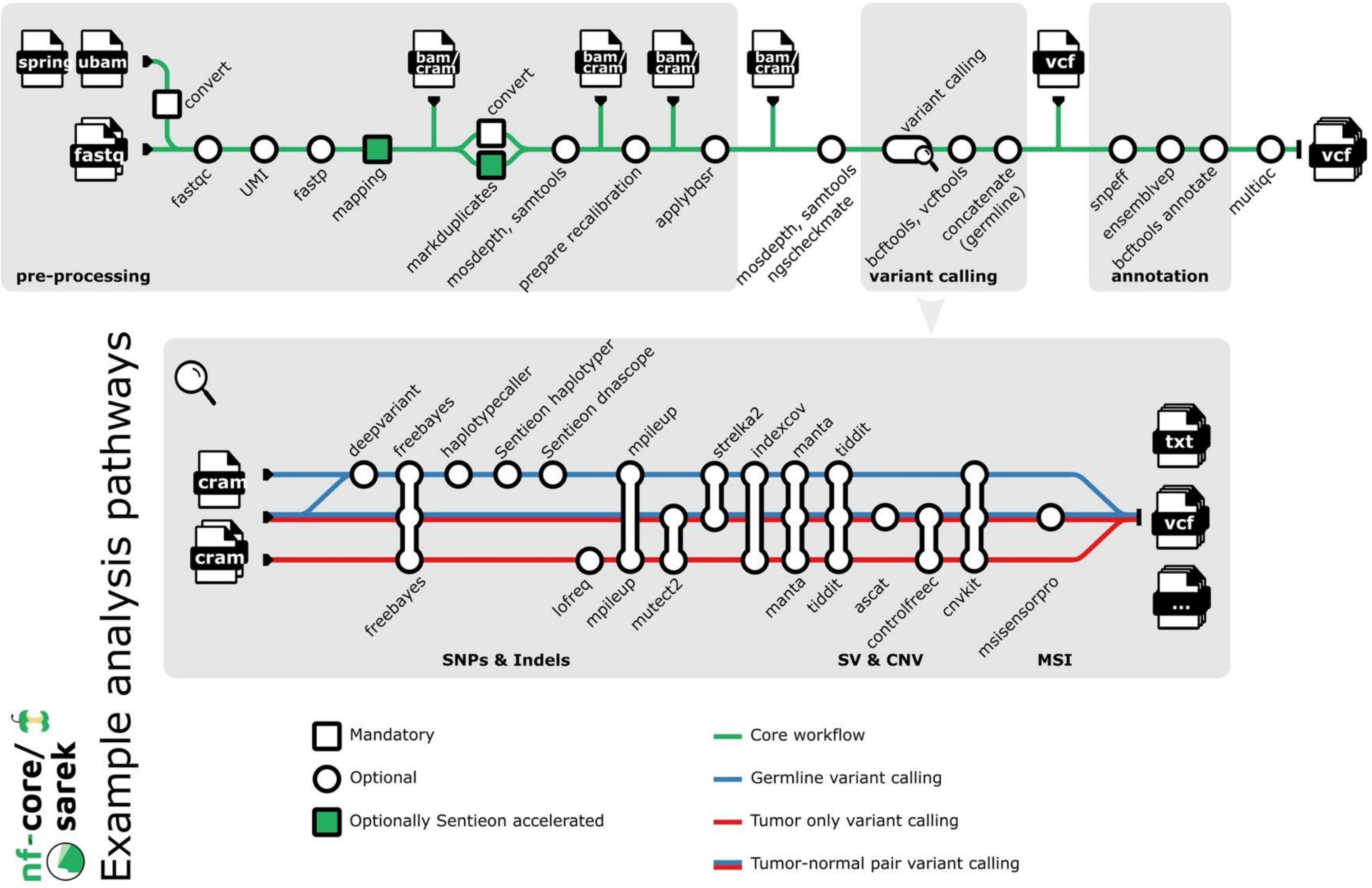




Clinical workflow for tumor genome analysis 5



Somatic Variant Analysis using Nextflow 6



Adapted from: Fellows Yates, James A., et al. PeerJ 9 (2021).

Linkage Analysis 7

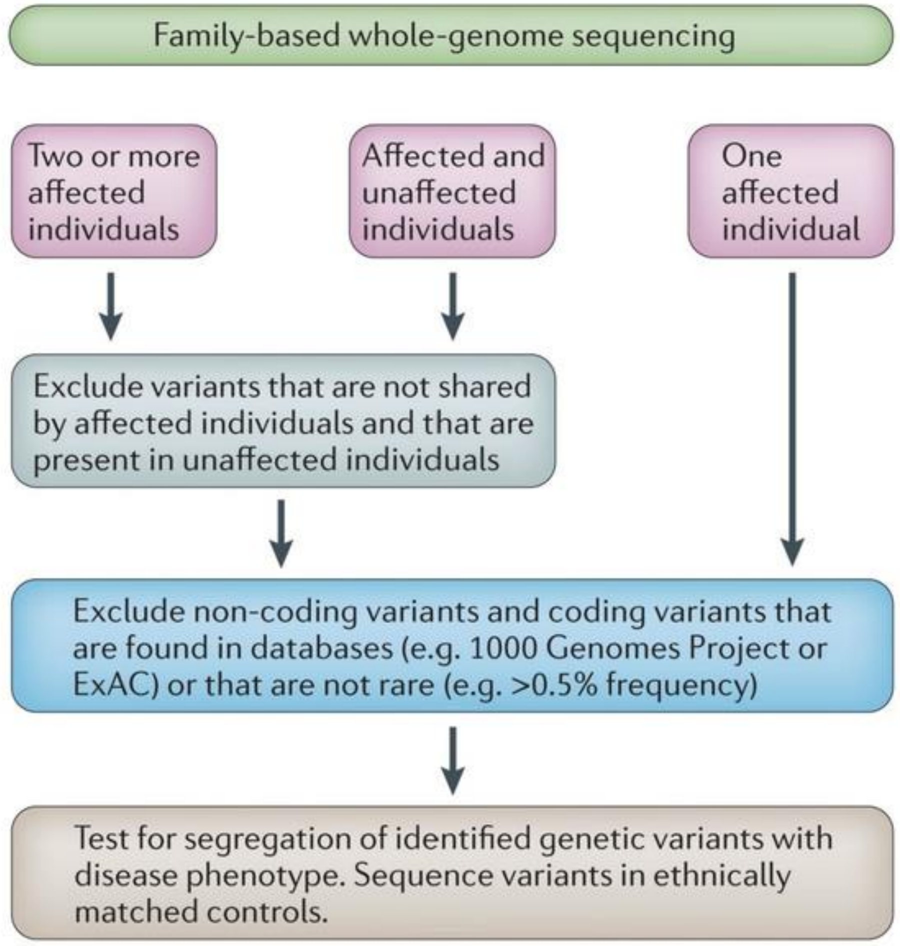
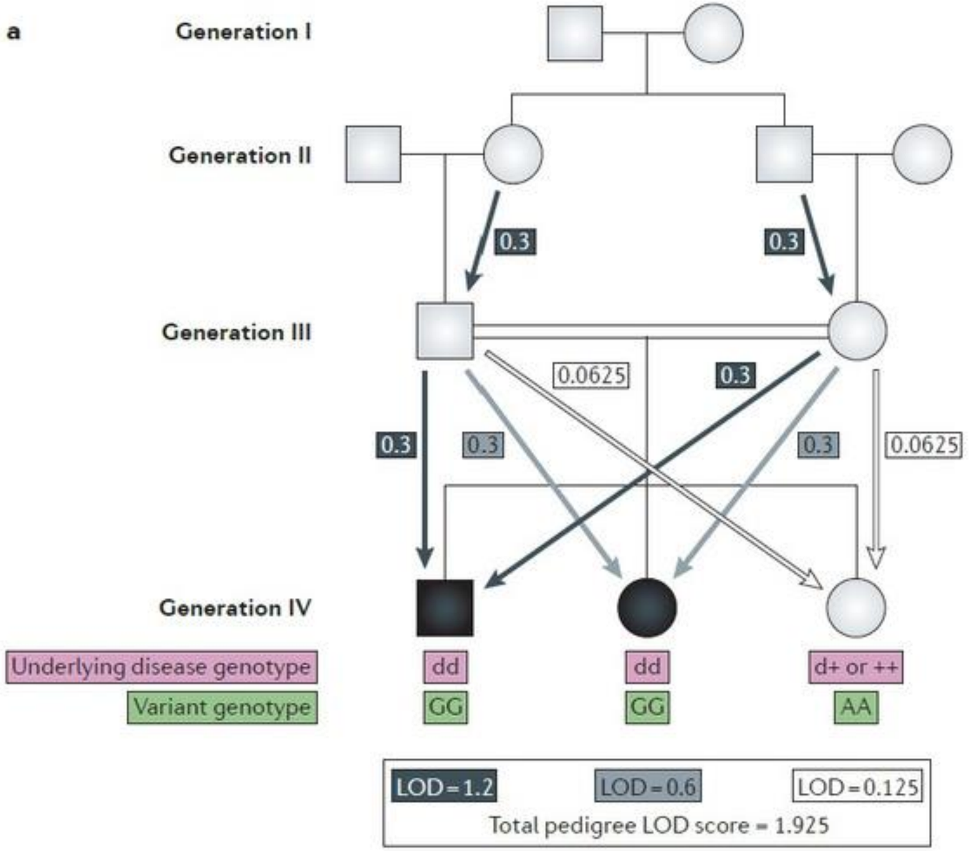
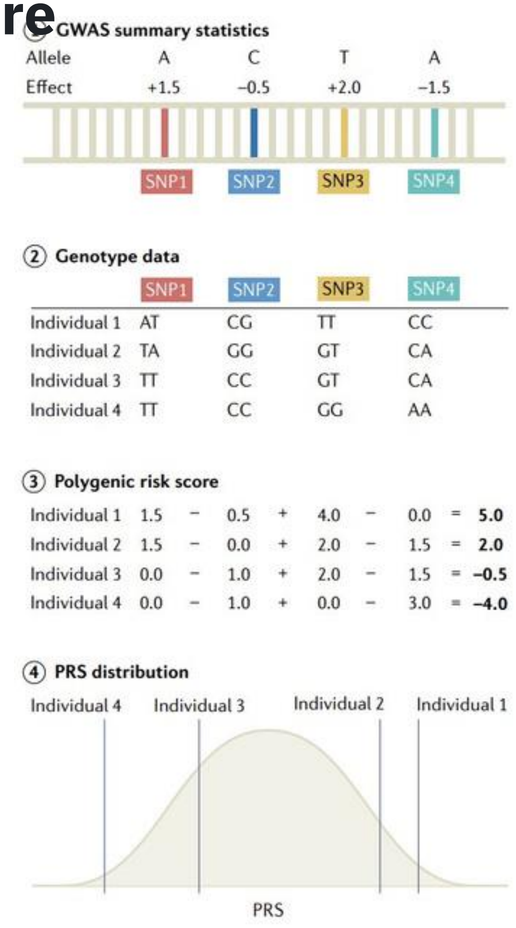


Figure 1. Workflow for the whole-genome sequencing filtering approach in human family data

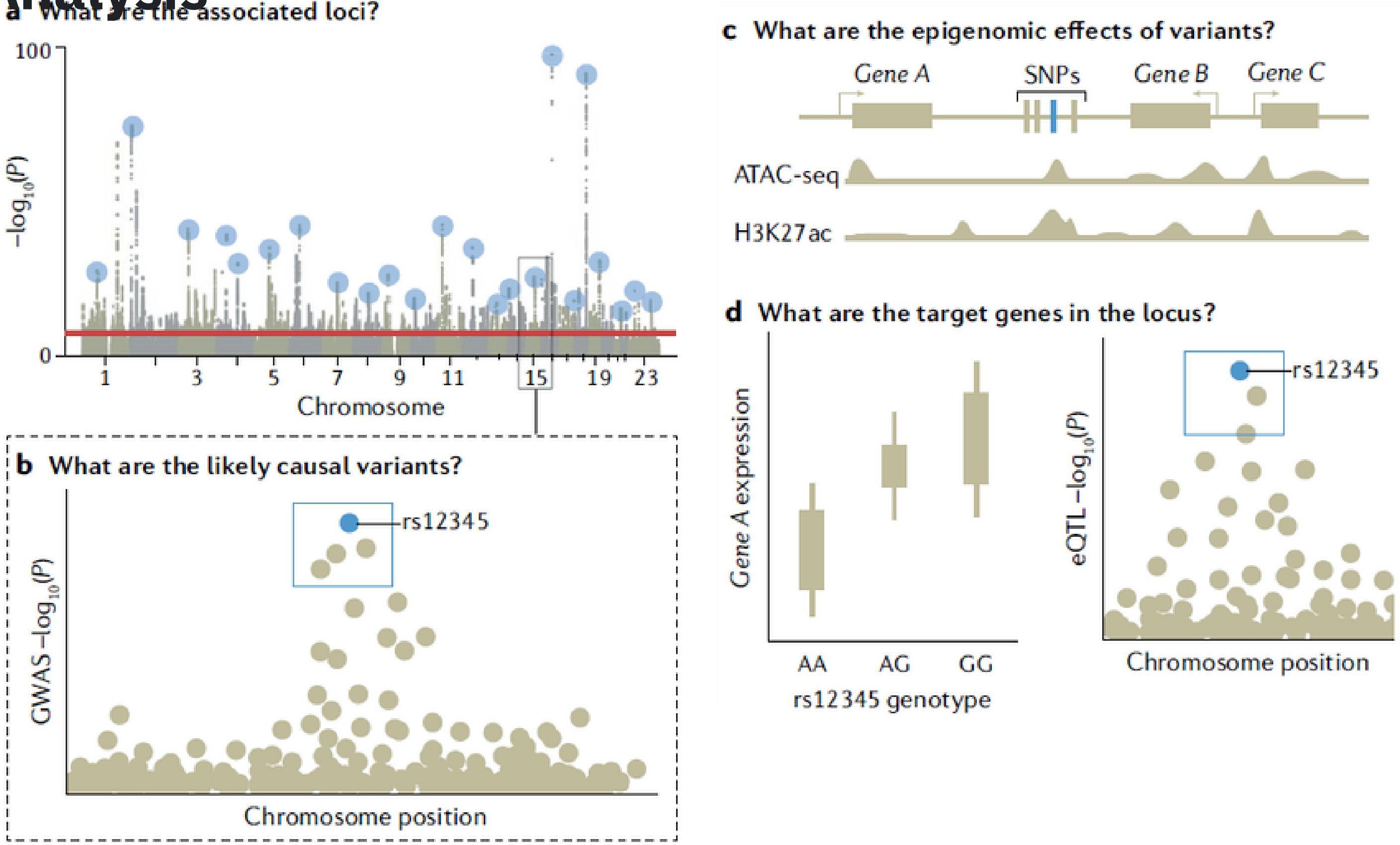


GWAS 8

(1) Polygenic Risk Score



(2) Functional Analysis



Class Survey (1%)

[What do want to learn from this course?]